5. What operator tests column for absence of data.

   a. NOT Operator
   b. Exists Operator
   c. IS NULL Operator
   d. IS UNIQUE operator
   e. None of the above

6. A deletion anomaly occurs when:

   a. Deleting a record from a table causes unintended changes in other tables.
   b. Deleting a record from a table requires deleting duplicate data.
   c. Deleting a record from a table results in the loss of other related data.
   d. Deleting a record from a table causes the loss of all data in that table.
   e. Deleting a record from a table causes the loss of all related data in other tables.

7. Which of the following statements about anomalies in database normalization is true?

   a. Anomalies occur only in unnormalized databases.
   b. Anomalies can lead to inconsistencies and errors in data management.
   c. Normalization eliminates all types of anomalies from a database.
   d. Anomalies are always caused by data redundancy.
   e. Anomalies are only a concern in small-scale databases.

8. Which of the following is a violation of the first normal form (1NF)?

   a. A relation with no duplicate tuples.
   b. A relation with no repeating groups.
   c. A relation with no multi-valued dependencies.
   d. A relation with no functional dependencies.
   e. A relation with attributes that contain multiple values.

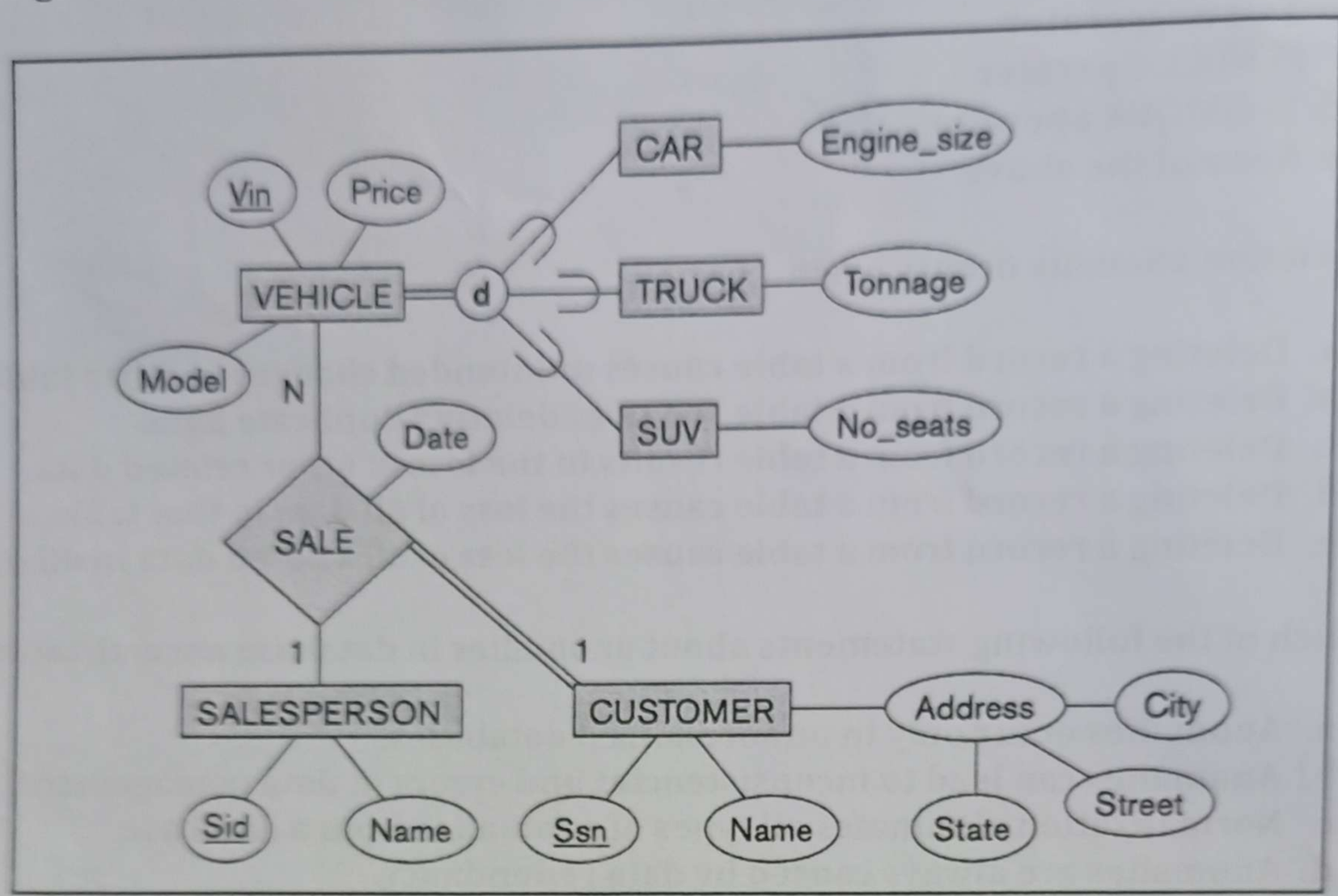
9. Which normal form is designed to eliminate insertion and deletion anomalies?

   a. First Normal Form (1NF)
   b. Second Normal Form (2NF)
   c. Third Normal Form (3NF)
   d. Fourth Normal Form (4NF)
   e. Fifth Normal Form (5NF)

10. In the context of normalization, why is it important to address partial functional dependency?

   a. To eliminate all redundancy from the database.
   b. To ensure that every attribute is dependent on the primary key.
   c. To allow multi-valued dependencies in a relation.
   d. To eliminate dependencies on parts of the primary key.
   e. None of the above.

Part B:
Consider the given EER diagram and answer the following questions.



11. A customer wants to purchase a vehicle with a specific engine size. Which path through the EER diagram would allow you to find the appropriate records?

a. CUSTOMER → SALE → VEHICLE → CAR
b. CUSTOMER → SALE → CAR
c. SALE → CAR
d. VEHICLE → CAR → Engine_size

12. In the diagram, if a new entity MOTORCYCLE is to be added with attributes specific to motorcycles only, how should it be linked to the existing entities?

a. As a separate entity with a foreign key to the VEHICLE entity
b. As a specialization of the VEHICLE entity, similar to CAR, TRUCK, and SUV
c. As a new standalone entity without any link to the VEHICLE entity
d. By adding its specific attributes to the VEHICLE entity and disregarding specialization

13. Which of the following is not a full functional dependency?

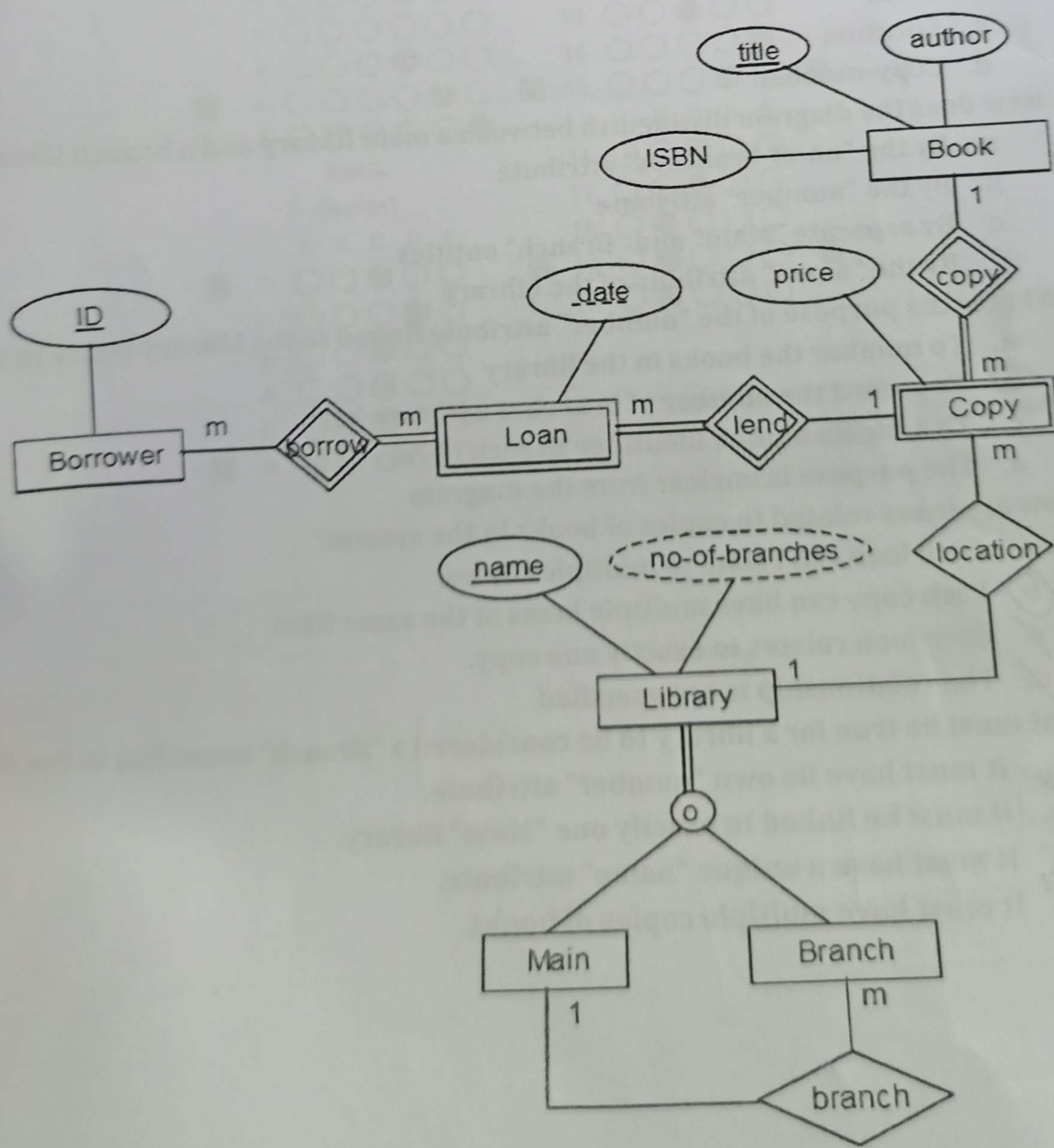
a. Sid → Name
b. Vin → Price, Model
c. Ssn, Name → City
d. None of the above
e. All of the above

Part C:

A library service wants to create a database to store details of its libraries, books and borrowers. Details include the following:

A book has a unique ISBN number, a title and one or more authors. The library service may own several copies of a given book, each of which is located in one of the service's libraries. A given library contains many books, and in order to distinguish different copies of the same book a library assigns a different copy-number to each of its copies of a given book; the price that was paid for each copy is also recorded. Every library has a unique name and is either a main library or a branch library. A main library may have zero or more branch libraries and every branch library is a branch of exactly one main library. A borrower has a name and a unique ID code. A borrower can have many books on loan, but each copy of a book can only be on loan to one borrower. A borrower could borrow the same book on several occasions, but it is assumed that each such loan will take place on a different date.

Below diagram illustrates a preliminary design of an Extended-Entity-Relationship model intended to meet the above specifications.



By using the above EER diagram answer the following mcqs.

14. What is the error with the Main and Branch entities in the provided EER model?
    a. There is no error; they are correctly modeled.
    b. They should be combined into one Library entity.
    c. Each Branch should be an attribute of Main.
    d. The number of branches should not be mentioned.
    e. The specialization should be disjoint
15. How is the relationship between a Copy and a Loan incorrectly depicted in the provided EER model?
    a. It shows a many-to-many relationship.
    b. It does not allow for multiple loans of the same copy.
    c. It correctly depicts a one-to-one relationship.
    d. It omits the possibility of renewing loans.
16. Which attribute is unnecessarily duplicated for the Copy entity in the provided EER model?
    a. ISBN
    b. Price
    c. Location
    d. Copy-number
17. How does the diagram distinguish between a main library and a branch library?
    a. By the "no-of-branches" attribute
    b. By the "number" attribute
    c. By separate "Main" and "Branch" entities
    d. By the "name" attribute of the Library
18. What is the purpose of the "number" attribute linked to the Library entity in the diagram?
    a. To number the books in the library
    b. To record the number of branches a library has
    c. To assign a unique identifier to each library
    d. The purpose is unclear from the diagram
19. How are loans related to copies of books in the system?
    a. Each loan can relate to multiple copies.
    b. Each copy can have multiple loans at the same time.
    c. Each loan relates to exactly one copy.
    d. The relationship is not specified.
20. What must be true for a library to be considered a "Branch" according to the diagram?
    a. It must have its own "number" attribute.
    b. It must be linked to exactly one "Main" library.
    c. It must have a unique "name" attribute.
    d. It must have multiple copies of books.

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Fill the following sheet to answer question 1. Select the correct choice by filling the circle completely.

NAME :	
EXAM :	
DATE :	

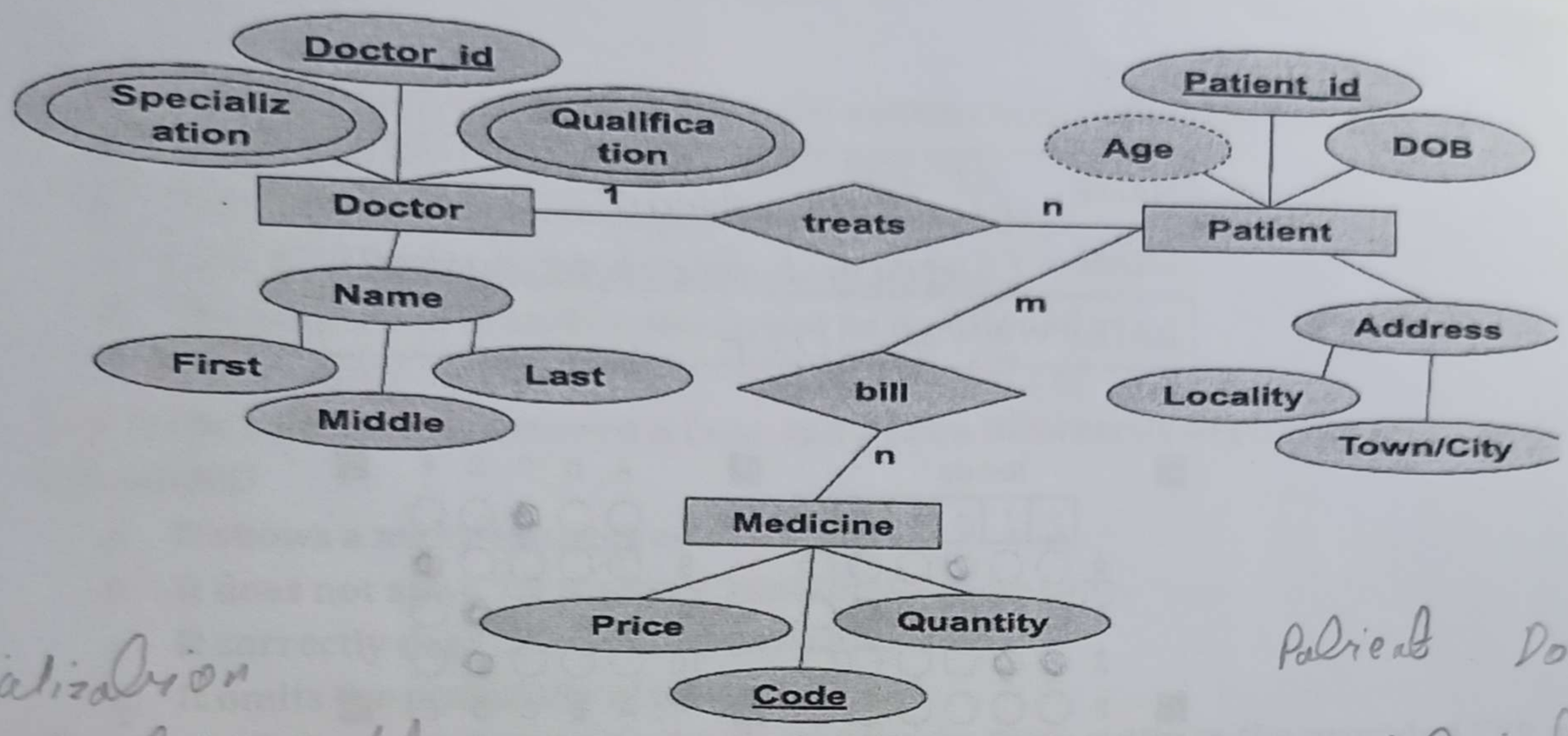
Roll No

Part A

Section1

	A	B	C	D	E
1	○	○	●	○	○
2	○	○	○	●	○
3	○	○	●	○	○
4	○	○	●	○	○
5	○	○	●	○	○
6	●	○	○	○	○
7	○	○	●	○	○
8	○	○	○	○	●
9	○	○	○	●	○
10	○	○	○	●	○
11	●	○	○	○	○
12	○	●	○	○	○
13	○	○	●	○	○
14	○	○	○	○	●
15	○	○	○	●	○
16	○	○	●	○	○
17	○	○	●	○	○
18	○	●	○	○	○
19	○	●	○	○	○
20	○	●	○	○	○

Question 2 [10 Marks]



Doctor
specialization
Qualification table

Patient Doctor ID
Patient id - Bill
code ~~Medicine~~ Quan Price
code Patient ID

Convert the above ERD into a relational model. Provide complete list of tables along with all attributes and constraints. Properly specify all primary and foreign key constraints.

∵ 1:n relation stated above

Total Tables: 6

Table : Patient → Primary key

Patient_ID	DOB	Doc_id	Local	Town

Table : Doctor

Doctor_id	FName	MName	LName

Primary key

Foreign key

Table : Specialization

Doctor_id	specialization

Composite Primary key

Foreign key

Table : Bill

Foreign key

Code	Patient_ID

Composite Primary key

Foreign key

Table : Qualification

Doctor_id	qualification

Composite Primary key

Foreign key

Table : Medicine

Code	Price	Quantity

Primary key

10

Question 3 [2*5=10 Marks]

The following case study models a mobile phone company. The company has a number of phones that are sold by staff to various clients. Each phone comes with a plan and each plan has a number of features specific to that plan including:

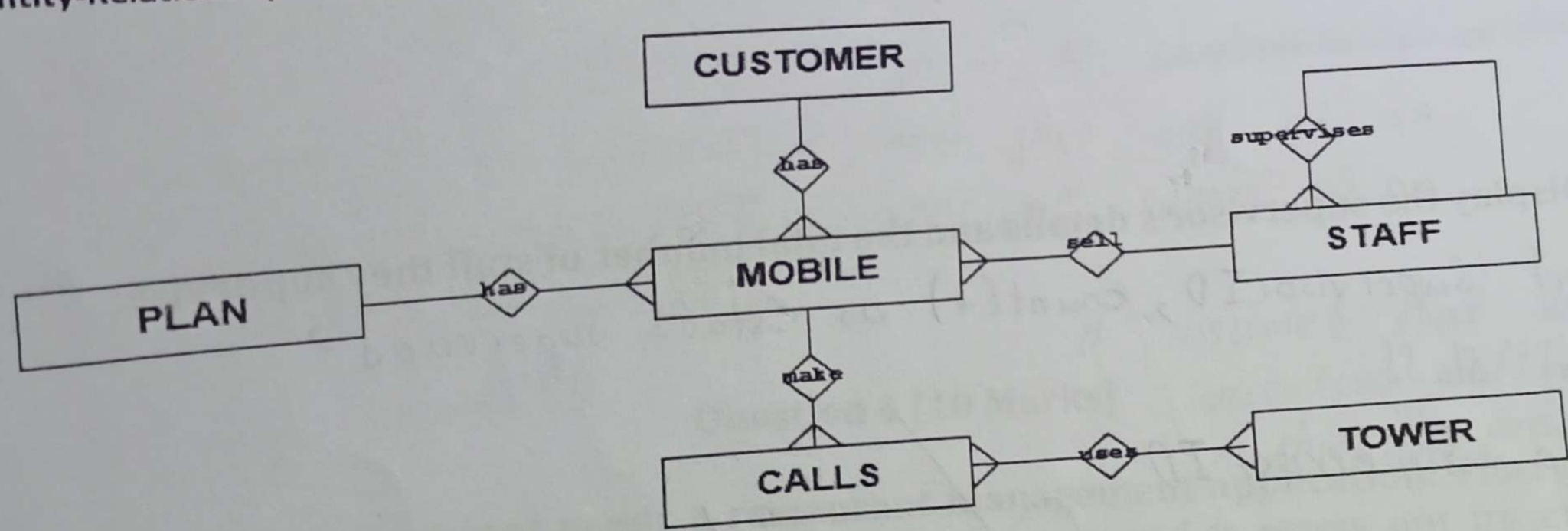
- A call charge in cents per minute (this does not apply to all plans).
- A plan duration in months.
- A break fee if the customer leaves the plan before the end of the plan duration.
- A monthly data allowance in gigabytes.

Assumptions that are made for the assignment are:

- Mobile phones are locked to a plan for the length of the plan duration.

Consider the following Entity-Relationship Diagram and Relational model:

Entity-Relationship Diagram



Relational Model

CUSTOMER	CALLS	MOBILE	STAFF	TOWER	CONNECT	PLAN
CustomerID	CallsID	MobileID	StaffID	TowerID	ConnectID	PlanName
Surname	MobileID	PhoneNumber	Surname	Location	TowerID	BreakFee
Given	CalledNumber	BrandName	Given	Bandwidth	CallsID	DataAllowance
DOB	CallDate	Joined	DOB	MaxConn		MonthlyFee
Sex	CallTime	Cancelled	Sex	SignalType		PlanDuration
PhoneHome	CallDuration	PlanName	Joined			CallCharge
PhoneWork	DataUsage	PhoneColour	Resigned			
PhoneFax		CustomerID	Address			
Address		StaffID	Suburb			
Suburb			Postcode			
State			Phone			
Postcode			SupervisorID			
			Commission			
			RatePerHour			

~~Customer cl inner join~~

customer cl inner join mobile m

phoneHome = OR OR

By using above entity relationship diagram and relational model provide answers by writing valid sql queries.

1. List all the customers who are not active. A customer should be considered not active if they have no active phone number registered under their name.

Select *
From Customer
Where PhoneHome IS NULL AND PhoneWork IS NULL AND PhoneFax IS NULL;

0

2. Display the supervisor's details and the total number of staff they supervise.

Select SupervisorID, count(*) as 'Staff Supervised'
From Staff
Group SupervisorID

0

3. Display the total income generated by the company in 2019 from call charges based on gender. Assume call duration is recorded in seconds and calls are charged per minute.

0

4. **Who is the youngest and oldest customer of postcode 3181, having an iPhone?**

```
select Top 1 *
From Customer, mobile
~~order by~~ where postcode = 3181 And BrandName Like 'IPhone'
order by DOB;
```

// gives oldest

```
select Top 1 *
From Customer, mobile
where postcode = 3181 And BrandName like 'IPhone'
order by DOB desc;
```

// gives youngest

5. **Which tower(s) were used by customer 20010 to make the first call?**

```
Select top 1 CustomerID, TowerID
From customer cl inner join mobile m on cl.customerID = m.customerID inner join calls c2 on
m.mobileID = c2.CallsID inner join connect c3 on
c2.calls ID = c3.calls ID
order by c2.calls ID
```

// assumed that each successive callID is greater in than the previous one.

Question 4 [10 Marks]

An owner of a restaurant needs a restaurant management application. That application needs to store data about the company's employees and it starts out by creating the following table of employees.

1. **Analyze the given unnormalized dataset.**
2. **Apply normalization techniques to bring the dataset to Third Normal Form (3NF).**
3. **Provide the normalized tables along with explanations for each normalization step.**

1. Name is being repeated, state-code,

EMPLOYEE_ID	NAME	JOB_CODE	JOB	STATE_CODE	HOME_STATE
E001	Alice	J01	Chef	26	Michigan
E001	Alice	J02	Waiter	26	Michigan
E002	Bob	J02	Waiter	56	Wyoming
E002	Bob	J03	Bartender	56	Wyoming
E003	Alice	J01	Chef	56	Wyoming

Question 1 [20 Marks]

Part A:

Answer the following twenty multiple choice question in the sheet provided at the end of this question.

1. Which SQL command(s) can be used to remove an object from the database?
   a. DELETE
   b. EDIT
   c. DROP
   d. TRUNCATE
   e. ROLLBACK

2. Suppose relation T1 (A, B) has n tuples and relation T2 (A, C) has m tuples. Consider the following SQL query.

   SELECT A, C FROM T1, T2;

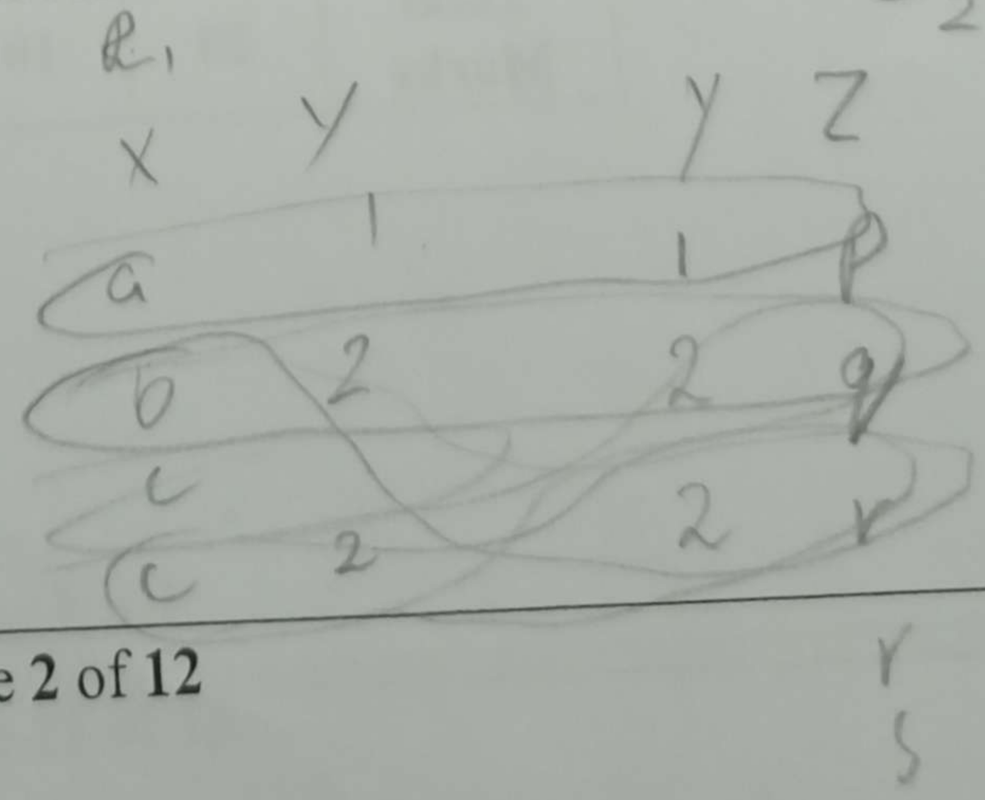
   What would be the number of tuples in the result of the above SQL query?
   a. N
   b. m
   c. n*m
   d. n+m
   e. Query gives an Error

3. Which data manipulation command is used to combines the records from one or more tables?

   a.select
   b.project
   c.join
   d.product
   e.merge

4. Suppose relation R1(X,Y) has tuples {(a,1), (b,2), (c, NULL), (c,2)} and relation R2(Y,Z) has tuples {(1,p),(2,q), (2,r), (NULL, r),(NULL, s)}.Consider the following SQL query.

   SELECT * FROM R1 LEFT OUTER JOIN R2 ON R1.Y = R2.Y;

   What would be the number of tuples in the result of the above SQL query?

   a.4
   b.5
   c.6
   d.8
   e.9